

Recursive Offline Distillation via Epistemic Complexity Approximation

Anonymous ACL submission

Abstract

Distillation of reasoning chains from Large Language Models (LLMs) into their smaller siblings is a standard training approach. Frequently, researchers do not have full access to a teacher model and its internal representations, which limits the resulting performance. We propose a novel blueprint for efficient offline distillation that recursively uses lightweight epistemic and aleatoric complexity approximations for selective distillation. Specifically, for three open models (3-4B), at each epoch, we estimate epistemic uncertainty using a medium model (70-72B) as an aleatoric proxy and sample only a small subset of the reasoning chains for training. Our approach consistently outperforms naive distillation with comparable computational budgets (57.15 ± 1.01 vs 51.61 on average) and offers a much steeper learning curve. Additionally, we present a comparative analysis of methods to generate and normalize distillation traces. We publish our code and data¹.

1 Introduction

Modern LLMs come in different sizes, and their performance is often correlated with the number of parameters. The largest of them demonstrate exceptional results and span 1-2 trillion parameters. Further domain adaptation (Cheng et al., 2024; Lu et al., 2024) or even inference at this scale requires massive computational resources (Wan et al., 2024) unavailable to most practitioners. It prompts the need for capable smaller models (Chen and Varoquaux, 2026) that can be fine-tuned and deployed in resource-constrained environments.

A prominent approach for small model training is distillation from a larger teacher model (Hinton et al., 2015; Sanh et al., 2020; Xu et al., 2024). Many methods rely on having access to the online teacher (Agarwal et al., 2024; Jiao et al., 2020; Gu

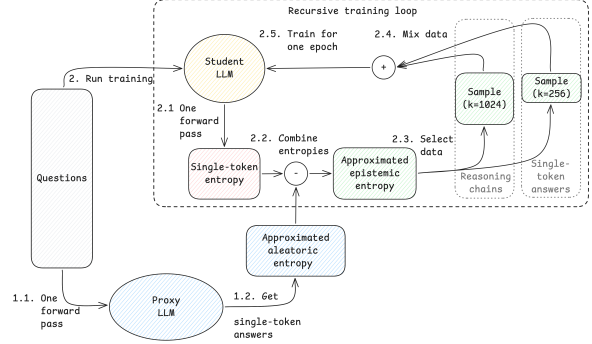


Figure 1: Recursive offline distillation scheme

et al., 2024), which is not always possible (Zhou et al., 2023; Ryskulov et al., 2026). It forces practitioners to explore offline distillation, where the student model is trained on the teacher’s pre-generated text output (Li et al., 2023; Hsieh et al., 2023).

We propose a fully automated recursive pipeline for offline distillation (Figure 1, described in 3) based on uncertainty decomposition (Wang et al., 2025b). The method consistently outperforms naive distillation (57.15 ± 1.01 vs 51.61 on average), and it is designed to work in resource-constrained environments.

Additionally, we provide a comparative analysis of augmentation strategies for synthetic reasoning chains when the teacher model does not perform perfectly in the target distillation domain and conduct sensitivity studies of the synthetic trace generation and normalization methods and sampling strategies.

2 Related Work

Recent work improved distillation through training objectives (Ko et al., 2024; Xu et al., 2026), data selection (Just et al., 2026; Li et al., 2025; Yang et al., 2026), and curriculum learning (Jiang et al., 2025; Zhang and Liu, 2026). Joint supervision with rationales and labels improves data efficiency (Hsieh et al., 2023). Curriculum benefits vary with

¹TODO

model capability, task complexity, and the choice of difficulty metric (Jia et al., 2026). ELAD (Zhang et al., 2024) selects uncertain student explanations for teacher revision. Complexity-aware fine-tuning (Goncharov et al., 2026) selects reasoning examples by single-token entropy. We recursively sample pre-generated traces using student-proxy entropy differences.

Trace synthesis includes explaining known solutions (Li and Mooney, 2024; Wang et al., 2025a) and learning from corrections (An et al., 2024; Yang et al., 2025). We compare direct generation and various augmentations on the same task under constrained trace lengths.

3 Method

Our central hypothesis is that we can approximate epistemic uncertainty of the student model with its entropy and entropy of a larger LLM. Goncharov et al. (2026) show that a single-token entropy (Figure 4) is a reasonable complexity predictor for a given entry in a question-answering dataset. We assume that it approximates the total uncertainty and that we can decompose it into epistemic and aleatoric components (Depeweg et al., 2018; Malinin and Gales, 2021). With scaling laws (Kaplan et al., 2020) in mind, we hypothesize that larger models should have lower total uncertainty. Therefore, in comparison to a small student model (3-4B), its substantially larger sibling (70-72B²) could be treated as a weak aleatoric proxy.

For a question x , assume $H(x) = A(x) + E(x)$, where H_S, H_P - single-token entropies (student and proxy), A - shared aleatoric component, and E_S, E_P - model-specific epistemic components. If its residual epistemic uncertainty satisfies $E_P(x) \leq \varepsilon(x)$, then

$$\begin{aligned} A(x) &= H_P(x) - E_P(x) \geq H_P(x) - \varepsilon(x) \\ E_S(x) &= H_S(x) - A(x) \leq H_S(x) - H_P(x) + \varepsilon(x) \end{aligned} \quad (1)$$

Thus, the entropy difference (*entropy gain*) approximates the student’s epistemic uncertainty with error at most $\varepsilon(x)$.

The proposed pipeline consists of the following steps (Figure 1, Algorithm 1): (1) prompt the larger model to answer all questions in the dataset with a single token once (or an ensemble, averaging entropies), collect entropy, assume

Algorithm 1 Recursive offline distillation via epistemic complexity approximation

Require: Student S , proxy model P , epochs T
Require: Questions D , ground-truth answers y_x , and pre-computed reasoning traces r_x
Ensure: Trained student S

```

1: for all  $x \in D$  do
2:    $H_P(x) \leftarrow \text{ENTROPY}(P, x)$ 
3: end for
4: for  $t = 1, \dots, T$  do
5:   for all  $x \in D$  do
6:      $H_S(x) \leftarrow \text{ENTROPY}(S, x)$ 
7:      $\text{EG}(x) \leftarrow \max\{H_S(x) - H_P(x), 0\}$ 
8:   end for
9:    $Q \leftarrow \text{WEIGHTEDSAMPLE}(D, \text{EG}, 1024)$ 
10:   $Q_A \leftarrow \text{first 256 questions in } Q$ 
11:   $B_R \leftarrow [(x, r_x) : x \in Q]$ 
12:   $B_A \leftarrow [(x, y_x) : x \in Q_A]$ 
13:   $S \leftarrow \text{TRAIN}(S, B_R \parallel B_A)$ 
14: end for
15: return  $S$ 

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it is an aleatoric uncertainty proxy; (2) for each training epoch, (2.1) prompt the student model to answer all questions in the dataset with a single token, collect entropy; (2.2) calculate entropy gain $\text{EG}(x) = \max\{H_S(x) - H_P(x), 0\}$; (2.3.1) sample 1024 synthetic reasoning chains without replacement, weighted by positive entropy gain; (2.3.2) sample 256 single-token answers similarly, as a subset of the previous sample; (2.4) concatenate samples; (2.5) train the student using LoRA (Hu et al., 2021) on reasoning traces first, followed by single-token responses.

4 Results

4.1 Experimental Setup

Dataset: Multiple-choice question-answering dataset MMLU Pro (Wang et al., 2024). It has questions of different complexities from 14 domains: ten options each, one correct option. Random train/test split - 80/20.

LLMs: Aleatoric proxy: ensemble of two medium models³ (averaging their entropies) - LLaMA 3.3 70B (Grattafiori et al., 2024), Qwen 2.5 72B (Qwen-Team, 2024). Students: Qwen2.5 3B (Qwen-Team, 2024), Phi-4-Mini (Microsoft et al., 2025), LLaMA 3.2 3B (Grattafiori et al., 2024). Reasoning trace source: DeepSeek V4 Pro and Flash (DeepSeek-AI et al., 2026) via API.

²This particular size is chosen based on the imposed resource constraints (80GB VRAM) and open model availability

³All models and datasets are published under permissive licenses that allow them to be used for research purposes

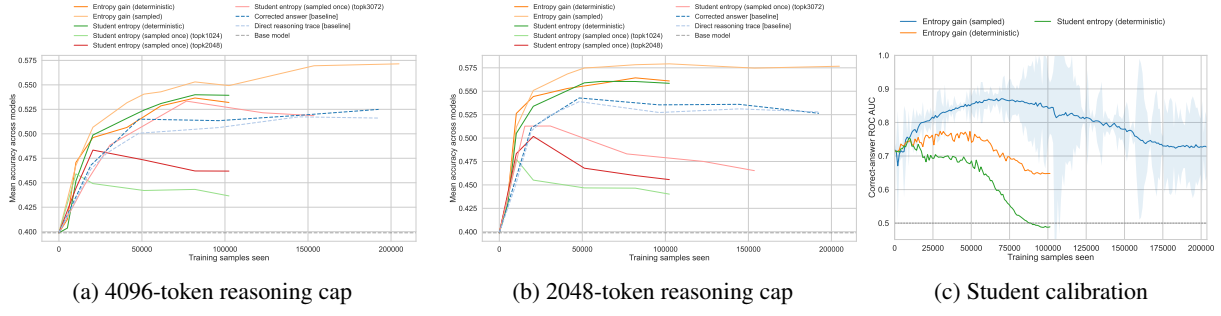


Figure 2: Distillation pipeline results across three models (average) on MMLU-Pro

Method	LLaMA 3.2 3B	Qwen 2.5 3B	Phi4-mini	Mean
Direct reasoning trace	46.09 / 45.55	51.45 / 50.37	57.27 / 62.43	51.61 / 52.78
Corrected answer	47.80 / 47.26	50.87 / 49.42	58.81 / 61.26	52.49 / 52.65
Student entropy (deterministic)	49.83 / 50.08	53.62 / 53.03	58.35 / 64.42	53.93 / 55.85
Entropy gain (deterministic)	<u>50.17 / 50.37</u>	51.04 / <u>53.28</u>	58.40 / 64.63	53.20 / <u>56.10</u>
Student entropy (sampled once, $k = 1024$)	41.35 / 40.11	39.44 / 39.07	50.17 / 52.87	43.65 / 44.01
Student entropy (sampled once, $k = 2048$)	42.19 / 41.06	43.18 / 43.02	53.20 / 52.62	46.19 / 45.57
Student entropy (sampled once, $k = 3072$)	42.02 / 42.81	56.77 / 40.86	56.77 / 55.90	51.86 / 46.52
Entropy gain (sampled, ours)	52.58 $\pm$ 1.94 / 52.08 $\pm$ 2.28	<u>56.64</u> $\pm$ 3.79 / 56.86 $\pm$ 2.32	62.22 $\pm$ 2.42 / 64.07 $\pm$ 4.52	57.15 $\pm$ 1.01 / 57.67 $\pm$ 1.19

Table 1: Distillation pipeline last-epoch accuracy (4096-token / 2,048-token thinking budget) on MMLU-Pro.

Hardware: We run all experiments⁴ on a single A100/H100⁵ GPU with 80GB of VRAM to validate the method in resource-constrained environments.

4.1.1 Our method and results

We evaluate the proposed pipeline for 200 epochs (205k questions seen) and 2 thinking budgets (2048 and 4096 tokens). It is compared under similar data budgets with:

Direct reasoning trace, *Corrected answer* - naive distillation on the most promising types of reasoning traces (see 4.3);

Student entropy (deterministic), *Entropy gain (deterministic)* - alternative pipelines with weighted sampling replaced with greedy sampling;

Student entropy (sampled once, $k=x$) - distillation executed against k samples selected once by student entropy (not recursively at each epoch).

Entropy gain (sampled), *Student entropy (deterministic)*, *Entropy gain (deterministic)* use the proposed data mix (reasoning traces/single-token answers = 1024/256); other methods use only reasoning traces.

Notably, the proposed data mix increases the number of training tokens by only 0.003% - 256 extra tokens compared to 8,388,608 tokens in 1024 reasoning chains.

All randomly sampled methods use 3 random

seeds, reporting $\pm 95\%$ confidence intervals, and deterministic methods are evaluated once with greedy sampling.

4.2 Main Results

Table 1 and Figure 2 show the results for MMLU Pro. We see that the proposed method produces a significant improvement over naive distillation (up to 10% gain). For both reasoning budgets, it provides the best result on average across three models in comparison with other approaches. Greedy sampling demonstrates subpar and less stable curves, though surpassing the baselines. Greedy distillation based on student entropy sampled once (Goncharov et al., 2026) shows signs of severe degradation after the initial uplift. However, Qwen 2.5 3B, notably, demonstrates the best performance with it at $k=3072$, but degrades at smaller sample sizes, revealing the fragility of the approach.

We also note that our method helps with the model calibration, significantly enhancing ROC AUC of single-token entropy for the first half of training. Shortly after, it starts declining, which is loosely correlated with the method’s plateau.

4.3 Synthetic Trace Generation

In resource-constrained environments, available VRAM imposes a strict limitation on the length of distilled reasoning chains. Once we start truncating them, we face a new challenge: which part of

⁴Training hyperparameters can be found in Appendix A.

⁵We had to switch cards depending on availability

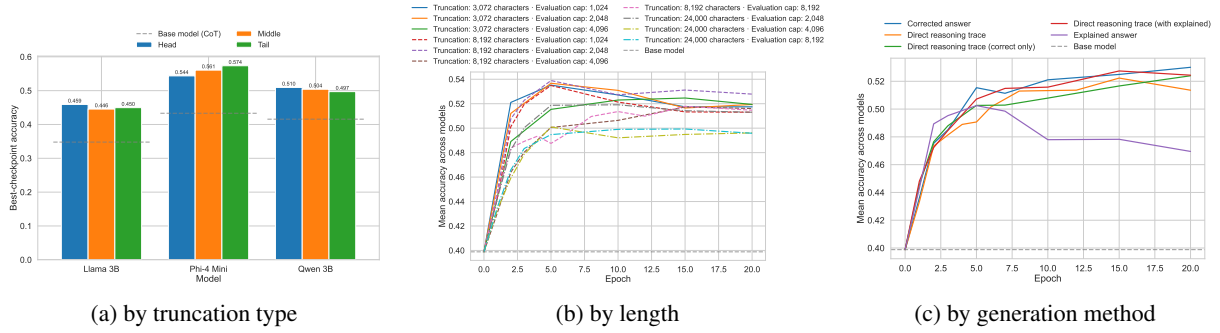


Figure 3: Comparison of synthetic traces across three models (average)

the trace contains the most useful signal? Finally, how exactly do we prompt the teacher model to generate reasoning traces?

We evaluate reasoning traces by: (a) *truncation type* - which part of the trace we keep when it exceeds the maximum length; (b) *length* - the optimal training truncation limit (characters) and evaluation thinking budget (tokens); (c) *generation method* (below).

Generation methods (details in Appendix C): (1) *Direct reasoning trace* - baseline prompt to just answer the question; (2) *Direct reasoning trace (correct only)* - same as (1), but we keep only traces with correct answers; (3) *Explained answer* - we give a correct answer to the model and prompt it to explain it; (4) *Direct reasoning trace (with explained)* - same as (1), but replace incorrect traces with (3); (5) *Corrected answer* - same as (1), but for incorrect traces, we give the model correct answers and ask it to update its reasoning.

Figure 3 shows that models have a slight preference for head truncation, medium to short training chains and thinking budgets. Surprisingly, longer reasoning chains and greater thinking budgets lead to notable performance decreases. As to the generation methods, we see that augmented reasoning traces lead to elevated performance, while the consistent data leakage results in degradation. Based on the findings, we use 8,192-character head truncation and *Corrected answer* generation for the pipeline.

4.4 Sensitivity Study

To attribute performance gains to either the data mix or the entropy-based score and sampler, we compare the proposed pipeline (*entropy gain*) with: (1) alternative pipelines where we sample according to *student entropy* and *discounted entropy gain* (see below); (2) an alternative pipeline where we

Sampler	Three-model mean
Direct reasoning trace (baseline)	51.61
Random	55.08 ± 0.80
Student entropy	55.87 ± 1.40
Discounted entropy gain	<u>55.98</u>
Entropy gain	57.15 ± 1.01

Table 2: Distillation pipeline last-epoch accuracy for different sampling methods (4096-token thinking budget)

use a random sampler.

Discounted entropy gain is defined as $\max\{H_S(x) - k(t) * H_P(x), 0\}$, $k(t) \in [0, 1]$, where t is the training epoch. We set $k(t)$ initially to 0.8 and decrease it according to a schedule, hypothesizing that, as the student undergoes training, the static medium model stops being a good aleatoric proxy.

Table 2 shows that, across three models on average, entropy gain provides the best results, confirming the contribution of the aleatoric uncertainty approximation. At the same time, we notice that random sampling performs exceptionally well (confirming findings of Okanovic et al. (2023)) compared to the baseline, which points to the proposed data mix as a significant contributor to the gains.

5 Conclusion and Discussion

This paper introduces a recursive offline distillation method rooted in epistemic uncertainty approximation (57.15 ± 1.01 mean accuracy across three students on MMLU-Pro vs 51.61 for direct distillation) with a much steeper learning curve (baseline score at 15% of its data budget). We confirm that both the entropy difference sampler and the data mix contribute to the gains. Surprisingly, shorter traces and reasoning budgets perform better. Further exploration of more rigorous epistemic uncertainty estimation and the surprising effect of the small data mix augmentation remains future work.

Limitations

- The method is tested only with small models. We do not know how well it is going to generalize to larger LLMs. Additionally, it might be challenging to find a suitable aleatoric proxy for larger students.
- We use a single dataset. We do not know how well it translates to other settings.
- Alternative aleatoric approximations, such as ensemble agreement and others, are left unexplored.
- We did not test and compare multiple sizes of aleatoric proxies and their impact on performance.
- Different compositions of the data mix are not tested due to lack of resources.
- We tested only a single schedule for *Discounted Entropy Gain*. Alternatives remain to be explored.
- Some low-resource environments might have even more constraints and therefore struggle to fit a suitable aleatoric proxy into VRAM.

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A Experimental Hyperparameters

Hyperparameter	Value
Optimizer	AdamW (adamw_torch)
Learning rate	10^{-4}
Learning-rate schedule	Cosine
Warmup ratio	0.06
Weight decay	0
Adam (β_1, β_2)	(0.9, 0.999)
Adam epsilon	10^{-8}
Maximum gradient norm	1.0
Mixed precision	BF16 enabled
Gradient checkpointing	Enabled
torch.compile	Enabled
LoRA rank r	8
LoRA alpha	16
LoRA scaling α/r	2
LoRA dropout	0.1
Initialization	Gaussian

B Single-token entropy

Single-token entropy (Figure 4) measures a model’s uncertainty over the first answer token

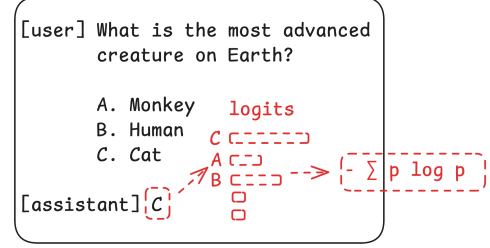


Figure 4: Single-token entropy example

when prompted to answer a question with a single token. For a question x , we apply softmax to the first-token logits to obtain probabilities $p(v | x)$ over the model’s full vocabulary V , and compute normalized Shannon entropy:

$$H(x) = -\frac{1}{\log |V|} \sum_{v \in V} p(v | x) \log p(v | x). \quad (2)$$

Dividing by $\log |V|$ puts student and proxy entropies on a common $[0, 1]$ scale despite differences in vocabulary size: values near zero indicate a concentrated prediction, while values near one indicate a nearly uniform distribution.

C Synthetic trace generation methods

We compare five ways to construct reasoning-and-answer targets for the same multiple-choice questions. Directly generated option letters are checked against ground-truth labels.

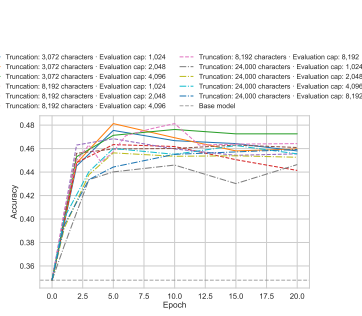
Direct reasoning trace. The teacher receives the question and options, generates reasoning, and predicts an option without seeing the ground-truth answer. The baseline retains both correct and incorrect responses.

Direct reasoning trace (correct only). We retain only baseline responses whose predicted option matches the ground truth, reducing the training set by discarding failed attempts.

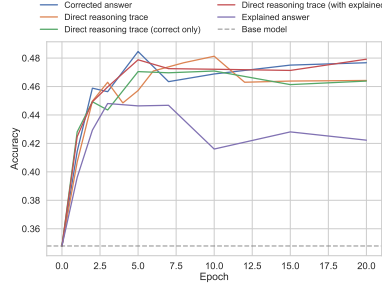
Explained answer. The teacher receives the question, options, and correct answer. It generates a step-by-step explanation that reads as a fresh solution, allowing alternatives and self-correction while avoiding acknowledgment of the supplied answer. The target pairs this explanation with the ground-truth option.

Direct reasoning trace (with explained). We retain correct baseline responses and replace each

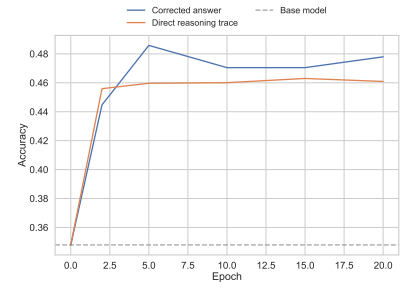
553	incorrect response with the explained-answer re-	User prompts. All three begin with:	604
554	sponse for the same question. This preserves ques-		
555	tion coverage while restricting answer-conditioned	Question: {question}	605
556	generation to failed attempts.	Options:	606
		a. {option text}	607
		b. {option text}	608
		...	609
557	Corrected answer. For each incorrect baseline	For explained answers, append <code>Correct option:</code>	610
558	response, the teacher receives the original reason-	followed by the ground-truth letter. For corrections,	611
559	ing, its incorrect option, and the correct option. It	append:	612
560	continues the trace toward the correct answer, pre-		
561	senting a natural self-correction without restarting	Partial reasoning so far:	613
562	or repeating the original text. We append the con-	{original reasoning}	614
563	tinuation and use the ground-truth option as the	Incorrect option this reasoning led to:	615
564	final answer. Correct baseline responses remain	{predicted letter}	616
565	intact.	Correct option: {ground-truth letter}	617
566	C.1 Prompt templates	C.2 Illustrative examples	618
567	The placeholders below are filled with the subject,	For brevity, consider a toy question with two op-	619
568	question, and options. Correct-only filtering and	tions: “A rectangle has sides of length 3 and 2.	620
569	replacement with explanations reuse these prompts;	What is its perimeter? a. 6; b. 10.” The following	621
570	they introduce no additional generation prompt.	illustrative targets show how the methods handle	622
		an incorrect attempt.	623
571	Direct generation: system prompt. Reasoning	Direct trace. “The perimeter is $3 \times 2 = 6$. An-	624
572	is collected separately from the teacher’s final an-	swer: a.” The baseline retains this response; the	625
573	swer.	correct-only variant discards it.	626
574	The following are multiple choice questions about	Explained answer. Given correct option b, a	627
575	{subject}. Choose a correct option letter. Answer	fresh explanation is: “There are two sides of each	628
576	with a single symbol. Do not print anything else.	length, so the perimeter is $2(3 + 2) = 10$. An-	629
		swer: b.” The with-explained variant substitutes	630
577	Explained answer: system prompt.	this response for the incorrect direct trace.	631
578	The following is a multiple choice question about	Corrected answer. Given the original reason-	632
579	{subject}. The correct answer is provided to you,	ing, incorrect option a, and correct option b, a	633
580	but your task is to produce a detailed chain-of-	continuation is: “That product gives the area.	634
581	thought with step-by-step reasoning that reads as	For the perimeter, I need to add all four sides:	635
582	if you were solving the question from scratch—	$3 + 2 + 3 + 2 = 10$. Answer: b.” The target retains	636
583	including any natural uncertainty, consideration	the original reasoning, appends this continuation,	637
584	of multiple options, or self-correction. Do not	and ends with b.	638
585	state or imply that the answer was given to you.		
586	End your response with Answer: <letter>.	C.3 Per-model results	639
587	Corrected answer: system prompt.	Figures 5–7 show test accuracy across training	640
588	The following are multiple choice questions about	epochs for each student. The left panels compare	641
589	{subject}. You will be shown a question, a partial	head-truncation limits of 3,072, 8,192, and 24,000	642
590	chain-of-thought that was previously produced	characters for direct reasoning traces; each legend	643
591	for it, the incorrect option that chain-of-thought	separately specifies the evaluation cap in tokens.	644
592	arrived at, and the correct option letter. Continue	The generation-method panels hold head trunca-	645
593	the partial chain-of-thought from where it left off	tion fixed at 8,192 characters and compare methods	646
594	so that it naturally arrives at the correct option.	at the stated evaluation cap. Dashed horizontal	647
595	Do not restart the reasoning, do not repeat what	lines and epoch-zero values use the base model’s	648
596	was already written, and do not acknowledge that	accuracy when prompted to answer with chain-of-	649
597	the previous attempt was wrong or that the correct	thought (base models are not originally reasoners),	650
598	answer was given to you. Write the continua-	shared across evaluation budgets.	651
599	tion as if you were the same reasoner noticing a		
600	mistake or new consideration mid-thought and re-		
601	vising course. In the end, answer with the correct		
602	option letter. End your response with Answer:		
603	<letter>.		



(a) Head-truncation comparison

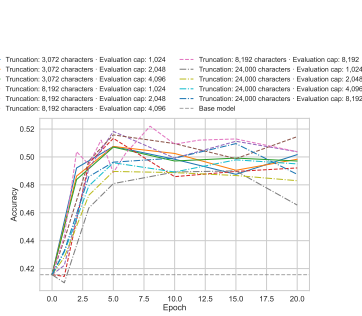


(b) Generation methods: truncation 8,192, eval cap 8,192

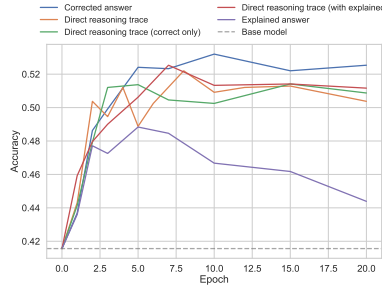


(c) Generation methods: truncation 8,192, eval cap 4,096

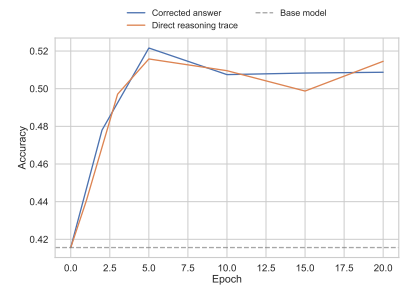
Figure 5: Synthetic-trace distillation results for LLaMA 3.2 3B.



(a) Head-truncation comparison



(b) Generation methods: truncation 8,192, eval cap 8,192



(c) Generation methods: truncation 8,192, eval cap 4,096

Figure 6: Synthetic-trace distillation results for Qwen 2.5 3B.

Budget (chars)	Direct traces	Corrected traces
3,072	3,603 (37.4%)	3,721 (38.7%)
8,192	1,879 (19.5%)	1,921 (20.0%)
24,000	738 (7.7%)	749 (7.8%)

Table 3: Number (percentage) of training traces exceeding each truncation budget

D Synthetic trace truncation and reasoning caps

Training truncation limits the retained teacher reasoning in characters; the reasoning cap limits the student’s generated reasoning at evaluation time in tokens. These are separate controls: a 3,072-character training trace can be evaluated with a 4,096-token cap. The final answer letter is retained separately when training traces are shortened.

Table 3 reports how many training traces exceed each character budget before truncation.

D.1 Which part of the trace is retained?

Head truncation keeps a prefix. Middle truncation keeps an initial segment and a final segment, dropping the interior of an overlong trace. Tail truncation keeps only a suffix. At the nominal 24,000-character setting, head retains up to 24,000

characters, middle retains the first 18,000 and last 6,000 of overlong traces, and tail retains the last 24,000. For corrected traces, middle truncation instead reserves up to 1,600 characters for the correction and uses the remaining space for the original prefix.

Figure 8 compares these policies at a fixed 8,192-token evaluation cap. Preferences vary by model: head gives the best observed checkpoint for LLaMA and Qwen, while tail does so for Phi4-mini.

D.2 Training truncation length versus evaluation cap

For direct traces with head truncation at 3,072, 8,192, or 24,000 characters, we compare available evaluation caps of 1,024, 2,048, 4,096, and 8,192 tokens. Figures 9 and 10 separate training length from evaluation budget. Best-checkpoint accuracy is the maximum over observed trained checkpoints for each cap; last-checkpoint accuracy uses the final available epoch. Only evaluated combinations are plotted.

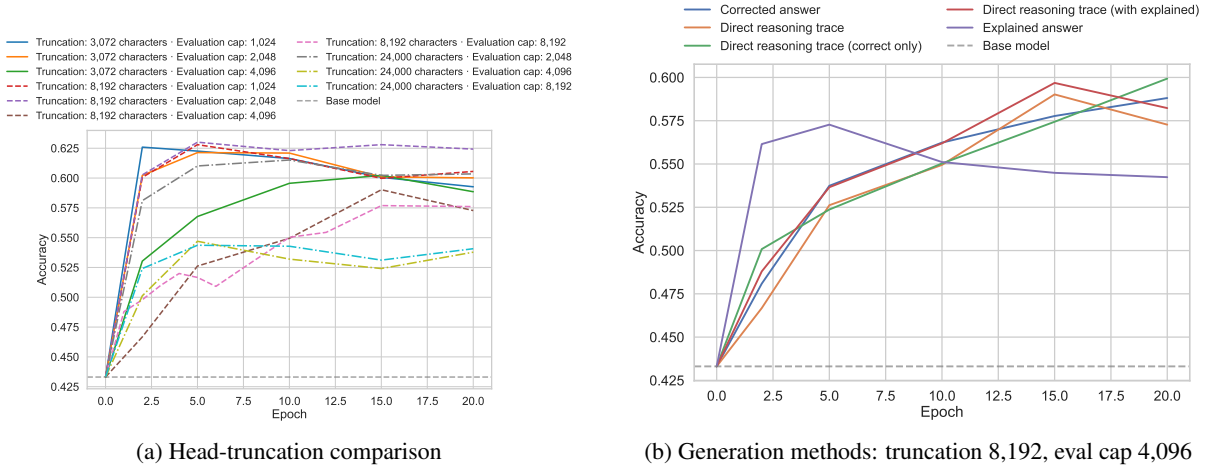


Figure 7: Synthetic-trace distillation results for Phi4-mini.

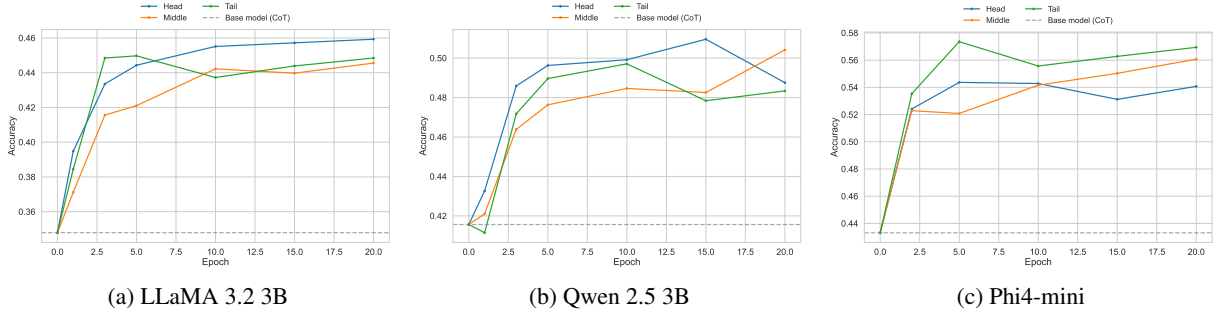


Figure 8: Training truncation policies at the nominal 24,000-character setting, evaluated with an 8,192-token reasoning cap. Dashed lines and epoch-zero points give the base model’s chain-of-thought accuracy.

E Sensitivity Study Details

do not increase it.

The sampler ablation holds the corrected traces and the per-epoch mix of 1,024 reasoning examples and 256 answer-only examples fixed, varying the selection rule: uniform random sampling or sampling without replacement weighted by student entropy, positive entropy gain, or discounted entropy gain. The discounted score uses $k(t) = 0.8 \cdot 0.01^{t/100}$, where t is the resampling epoch; its proxy contribution falls to 0.008 at epoch 100. It is labelled “Combined entropy (sampled)” in the graphs. Table 4 reports final accuracy; Figure 14 compares the weighted samplers and random sampling.

Figures 11–13 provide individual accuracy and coverage curves. Lines average available seed replicas. Horizontal axes count reasoning-question selections, excluding the additional answer-only examples: 1,024 per standard epoch, k for a fixed- k run, and 9,626 for full-dataset baselines. Accuracy plots share each model’s initial chain-of-thought score across evaluation budgets.

Coverage counts distinct question IDs selected at least once, so repeated epochs and response formats

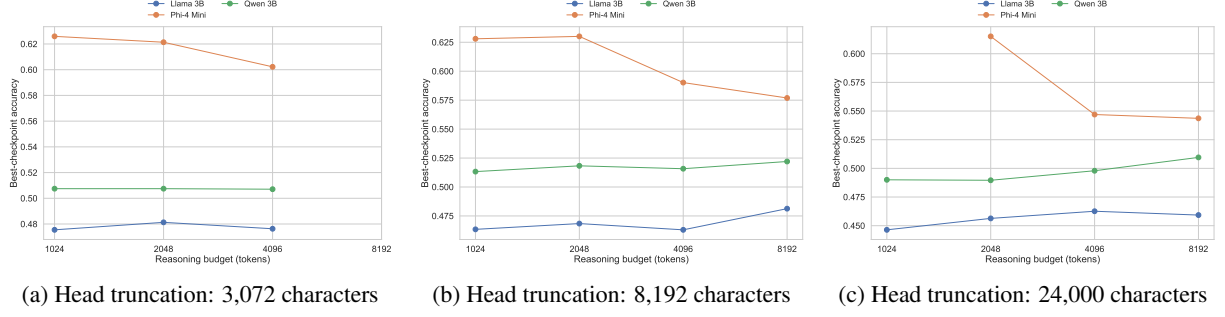


Figure 9: Best-checkpoint accuracy versus evaluation reasoning cap for each training truncation length.

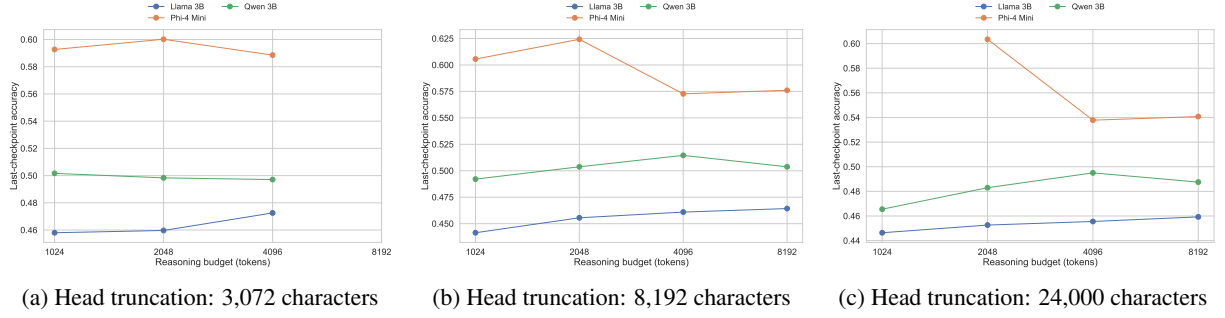


Figure 10: Last-checkpoint accuracy versus evaluation reasoning cap for each training truncation length.

Sampler	LLaMA 3.2 3B	Qwen 2.5 3B	Phi4-mini	Mean
Direct reasoning trace (baseline)	46.09	51.45	57.27	51.61
Random	51.77 $\pm$ 2.76	57.91 $\pm$ 1.40	55.57 $\pm$ 2.54	55.08 $\pm$ 0.80
Student entropy	51.18 $\pm$ 3.11	57.32 $\pm$ 1.52	59.12 $\pm$ 5.22	55.87 $\pm$ 1.40
Discounted entropy gain	51.50	56.90	<u>59.53</u> $\pm$ 2.11	<u>55.98</u>
Entropy gain	52.58 $\pm$ 1.94	56.64 $\pm$ 3.79	62.22 $\pm$ 2.42	57.15 $\pm$ 1.01

Table 4: Distillation pipeline last-epoch accuracy for different sampling methods (4096-token thinking budget)

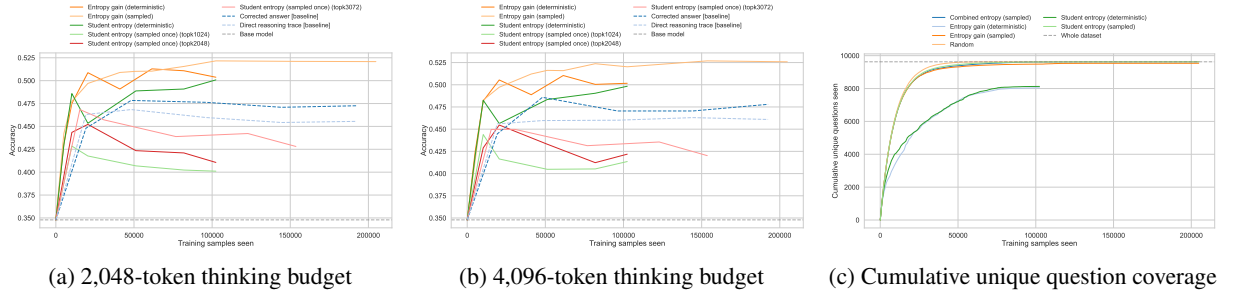


Figure 11: Distillation by sampling strategy for LLaMA 3.2 3B.

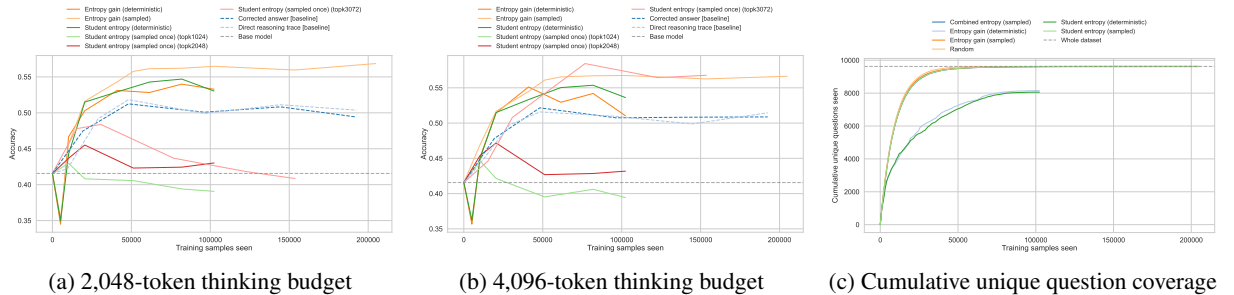


Figure 12: Distillation by sampling strategy for Qwen 2.5 3B.

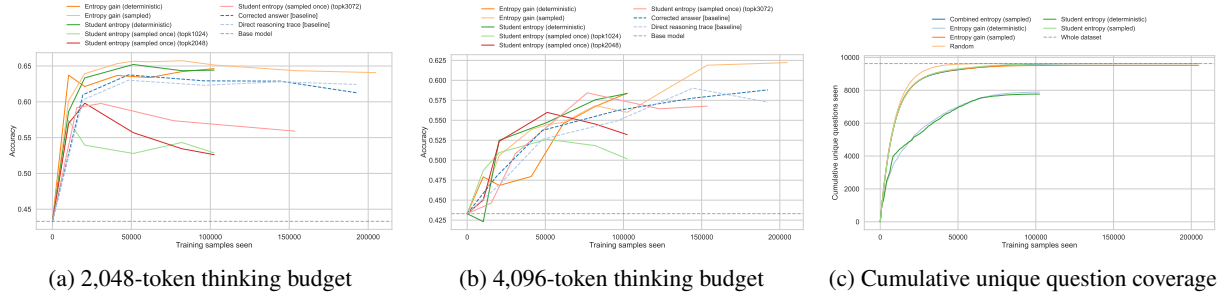


Figure 13: Distillation by sampling strategy for Phi4-mini.

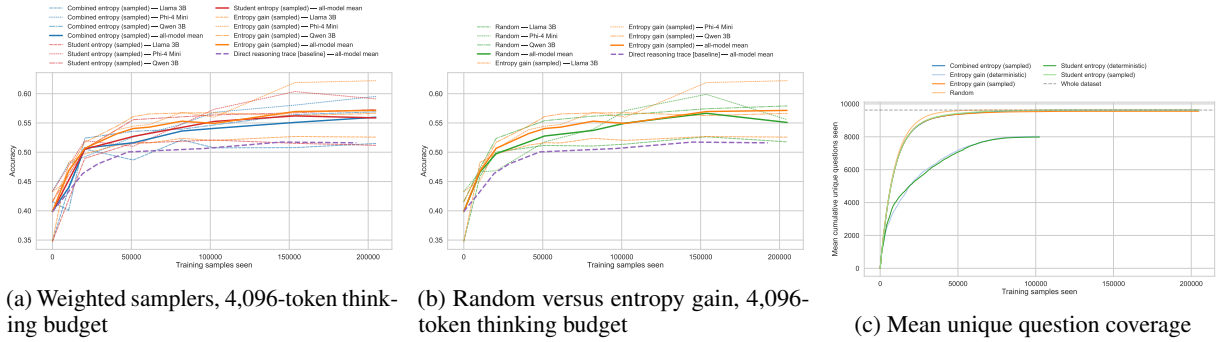


Figure 14: Sampler comparisons and question coverage across all three models. In the accuracy panels, thin dashed curves show individual models, thick solid curves their equal-weight means, and the thick dashed curve the direct-trace baseline mean.